

Prompt Engineering Strategy Benchmark for RAG Pipelines

A systematic comparison of Zero-shot, Few-shot, Chain-of-Thought, and Structured Output prompting within a controlled Retrieval-Augmented Generation pipeline.

1 PROBLEM STATEMENT

Large Language Models often generate inaccurate or hallucinated responses when answering questions solely from internal knowledge. Retrieval-Augmented Generation (RAG) mitigates this by retrieving relevant external documents before answer generation — but answer quality also depends heavily on the prompting strategy used. This project systematically benchmarks four prompting strategies while keeping every other pipeline component constant, evaluating each on identical datasets, retrieval settings, embedding models, vector databases, and LLM configurations to identify the most reliable, grounded approach for production RAG systems.

- Zero-shot
- Few-shot
- Chain-of-Thought
- Structured Output

2 OBJECTIVES

- Build a common, reusable RAG pipeline as the fixed experimental baseline.
- Implement four distinct prompting strategies on top of it.
- Evaluate all strategies using an identical dataset and metric suite.
- Compare trade-offs in accuracy, hallucination rate, and response quality.

3 METHODOLOGY — RAG EVALUATION PIPELINE



4 TECHNOLOGY STACK

Python	LangChain / LlamaIndex
FAISS / ChromaDB	BGE / OpenAI Embeddings
Gemini / Llama / Qwen / OpenAI	RAGAS & Custom Metrics

5 EXPECTED OUTCOME

A comparative benchmark identifying the strengths and weaknesses of each prompting strategy for RAG pipelines, with practical recommendations for selecting prompting techniques in production AI applications.

EVALUATION METRICS

Accuracy

Hallucination Rate

Faithfulness

Precision

Recall

Relevance